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$$\begin{aligned}& \int_{-1}^1 \frac{dx}{x^2} \\& \int \frac{1}{x^2} dx = -\frac{1}{x} + c \\& \Rightarrow \int_{-1}^1 \frac{dx}{x^2} = \int_{-1}^0 \frac{dx}{x^2} + \int_0^1 \frac{dx}{x^2} \\& = \lim_{a \rightarrow 0-} \int_{-1}^a \frac{dx}{x^2} + \lim_{a \rightarrow 0+} \int_a^1 \frac{dx}{x^2} \\& = \lim_{a \rightarrow 0-} \left[-\frac{1}{x} \right]_{-1}^a + \lim_{a \rightarrow 0+} \left[-\frac{1}{x} \right]_a^1 \\& = \lim_{a \rightarrow 0-} \left(-\frac{1}{a} + \frac{1}{-1} \right) + \lim_{a \rightarrow 0+} \left(-\frac{1}{1} + \frac{1}{a} \right) \\& = \lim_{a \rightarrow 0-} \left(-\frac{1}{a} - 1 \right) + \lim_{a \rightarrow 0+} \left(-1 + \frac{1}{a} \right) \\& = (-(-\infty) - 1) + (-1 + (+\infty)) \\& = +\infty + (+\infty) \\& = +\infty \text{ (divergent, integraal bestaat niet)}\end{aligned}$$

References